

Deadline 23rd March 2025

Exercise: Power performance verification

Introduction to assignment AS03

Power performance verification

Identify valid inflow sector

Data filtering, data normalization & data binning

Determine measured, normalized power curve & uncertainty

Calculation of annual energy production (AEP) for different annual average wind speeds

Test site



Test site

	WGS84, UTM33		Dimensions	
	E [m]	N [m]	Hub height [m]	Diameter [m]
Mast (MM1)	317,438	6,175,028		
Turbine under test V52	317,548	6,174,985	44	52
Neighboring turbine V27	317,519	6,174,901	31.5	27

Identify valid undisturbed inflow sector

- Assessment of influences caused by wind turbines and obstacles at the test site
- Assessment of terrain at the site

Hints:

- “free sector” assessment:
 1. Obtain waked sectors from turbines
 2. Obtain waked sectors from relevant obstacles

(you can assume Risø campus building heights are aprox. 8-10m.)

Additionally, you may:

Look at a) Roughness changes vs direction(GoogleEarth); b) turbulence intensity vs direction ...

Would you reduce the measurement sector?

- “terrain assessment”: simplification seen in class (slope)

Input

Name	Unit	Height[m]	Description
date_time	-	-	Runname = timestamp with reference to the starting time for the 10 minute period.
ActPow	kW	-	Mean active power
ActPow_stdev	kW	-	Stdev of active power
ActPow_min	kW	-	Minimum active power
ActPow_max	kW	-	Maximum active power
ROT	rpm	-	Mean rotor speed
ROT_stdev	rpm	-	Stdev of rotor speed
ROT_min	rpm	-	Minimum rotor speed
ROT_max	rpm	-	Maximum rotor speed
Pitch	deg	-	Mean pitch angle
Pitch_stdev	deg	-	Stdev of pitch angle
Pitch_min	deg	-	Minimum pitch angle
Pitch_max	deg	-	Maximum pitch angle
yaw	deg	-	Mean yaw angle
yaw_stdev	deg	-	Stdev of yaw angle
Wsp_44m	m/s	44	Mean wind speed. Mounted on South boom
Wsp_44m_stdev	m/s	44	Stdev of wind speed. Mounted on South boom
TI_44m	%	44	Mean turbulence intensity. Mounted on South boom
Wdir_41m	deg	41	Mean wind direction. Mounted on North boom
Wdir_41m_stdev	deg	41	Stdev wind direction. Mounted on North boom
AirAbs_70m	degC	70	Mean temperature. Mounted on South boom
Press_enc_2m	hPa	2	Mean atm. pressure, measured in mast
RH_2m	%	2	Mean relative humidity

Annex1: Available 10-minute statistics in data file

Assume Direction, Temp.,
Press, Hum represent hub
height

Data normalization (1)

- Calculate air density $\rho_{10\text{min}}$:

$$\rho_{10\text{min}} = \frac{1}{T_{10\text{min}}} \left(\frac{B_{10\text{min}}}{R_0} - \Phi P_w \left(\frac{1}{R_0} - \frac{1}{R_w} \right) \right) \quad (12)$$

where

$\rho_{10\text{min}}$ is the derived 10 min averaged air density;

$T_{10\text{min}}$ is the measured absolute air temperature averaged over 10 min [K];

$B_{10\text{min}}$ is the air pressure corrected to hub height averaged over 10 min [Pa];

R_0 is the gas constant of dry air 287,05 [J/kgK];

Φ is the relative humidity (range ~~0 % to 100 %~~);

R_w is the gas constant of water vapour 461,5 [J/kgK];

P_w is the vapour pressure equal to $0,000\,020\,5 \exp(0,0631\,846\,T_{10\text{min}})$ [Pa];

Vapour pressure P_w depends on mean air temperature over the 10 min.

Data normalization (2)

- For stall regulated, constant rotor speed turbines the normalization is:

$$P_n = P_{10\min} \left(\frac{\rho_0}{\rho_{10\min}} \right)$$

where

P_n is the normalised power output;

$P_{10\min}$ is the measured power averaged over 10 min;

- For active controlled turbines the normalization is:

$$V_n = V_{10\min} \left(\frac{\rho_{10\min}}{\rho_0} \right)^{1/3}$$

where

V_n is the normalised wind speed;

$V_{10\min}$ is the measured wind speed averaged over 10 min.

Power curve determination

- Use wind speed bins: $\Delta=0.5$ m/s, centered around:
2.0, 2.5, 3.0, 3.5...m/s

$$V_i = \frac{1}{N_i} \sum_{j=1}^{N_i} V_{n,i,j}$$

$$P_i = \frac{1}{N_i} \sum_{j=1}^{N_i} P_{n,i,j}$$

where

V_i is the normalized and averaged wind speed in bin i ;

$V_{n,i,j}$ is the normalized wind speed of data set j in bin i ;

P_i is the normalized and averaged power output in bin i ;

$P_{n,i,j}$ is the normalized power output of data set j in bin i ;

N_i is the number of 10 min data sets in bin i .

Annual energy production (AEP)

- V_{AVE} = Rayleigh mean wind speed

$$AEP = N_h \sum_{i=1}^N [F(V_i) - F(V_{i-1})] \left(\frac{P_{i-1} + P_i}{2} \right)$$

where

AEP is the annual energy production;

N_h is the number of hours in one year $\approx 8\,760$;

N is the number of bins;

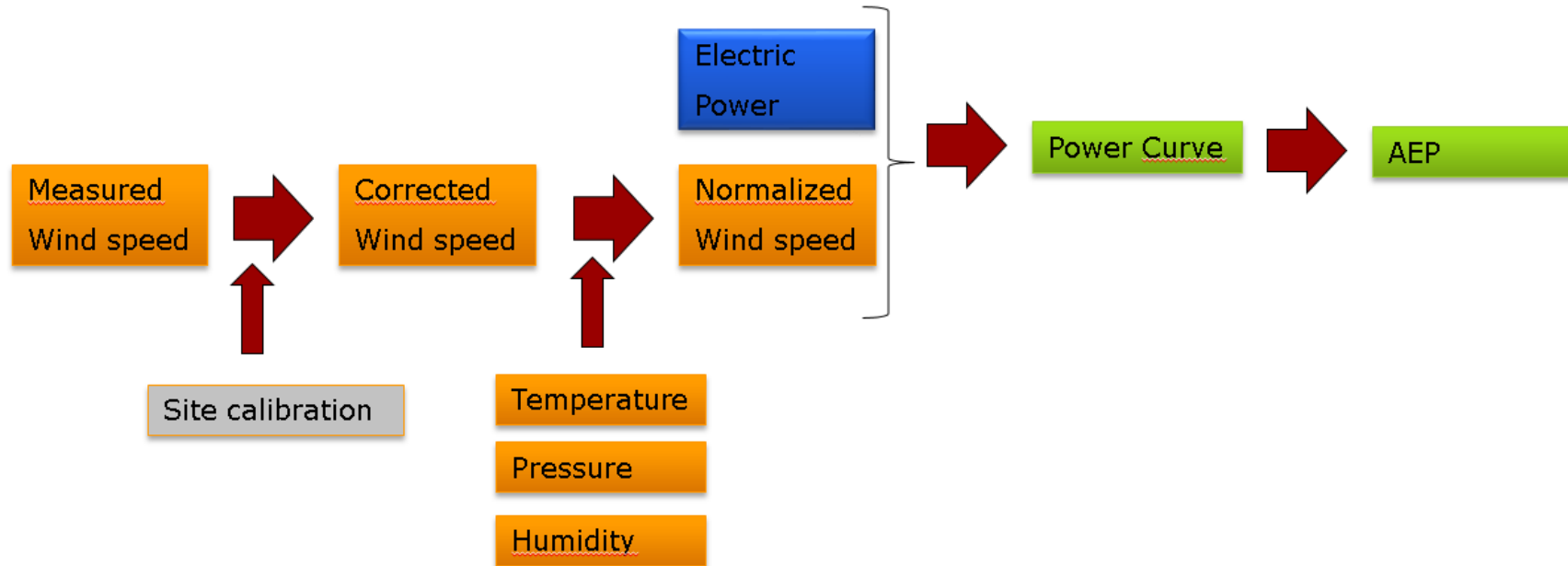
V_i is the normalized and averaged wind speed in bin i ;

P_i is the normalized and averaged power output in bin i .

$$F(V) = 1 - \exp\left(-\frac{\pi}{4} \left(\frac{V}{V_{ave}}\right)^2\right)$$

Calculation of AEP uncertainty (1)

- 1) Assess the **uncertainty** of each parameter (power, wind speed, temp, etc)
- 2) Derive **sensitivity factors**. I.e. "translate" all uncertainties to uncertainties in power
- 3) Combine uncertainties → Total uncertainty in PC and in AEP



Calculation of AEP uncertainty (2)

Evaluating the measurement uncertainty

- Combined uncertainty per power curve bin:

$$u_{c,i} = \sqrt{s_i^2 + u_i^2} \quad (\text{E.3})$$

- "Category A" uncertainty in the power:

$$s_i = s_{P,i} = \frac{\sigma_{P,i}}{\sqrt{N_i}} \quad (\text{E.9})$$

$\sigma_{P,i}$ ← standard deviation of the power data in bin i ;
 N_i ← the number of 10 min data sets in bin i ;

- Combination of all "Category B" uncertainty sources:

$$u_i = \sqrt{u_{P,i}^2 + c_{V,i}^2 u_{V,i}^2 + c_{T,i}^2 u_{T,i}^2 + c_{B,i}^2 u_{B,i}^2 + c_{RH,i}^2 u_{RH,i}^2} \quad (\text{E.63 corr.})$$

Calculation of AEP uncertainty (3)

Evaluating the measurement uncertainty

- AEP uncertainty:

$$u_{AEP} = N_h \sqrt{\sum_{i=1}^N f_i^2 s_i^2 + \left(\sum_{i=1}^N f_i u_i \right)^2} \quad (\text{E.66})$$

f_i is the relative occurrence of wind speed between V_{i-1} and V_i : $f_i = F(V_i) - F(V_{i-1})$ within bin i

Calculation of AEP uncertainty (4)

Annex 2: Parameters used to estimate the combined uncertainty of AEP

Item	Uncertainty, u	Sensitivity factor
Category B uncertainties in Electric power	$u_{P,i} = \sqrt{((0.002 \times P_i)^2 + 3.7^2 + 0.3^2)}$	1
Category B uncertainties in wind speed	$u_{V,i} = \sqrt{(0.025^2 + (0.038 + 0.0038 \times V_i)^2 + (0.01 \times V_i)^2 + (0.02 \times V_i)^2 + (0.001 \times V_i)^2)}$	$c_{V,i} = P_i - P_{i-1} / V_i - V_{i-1} $
Category B uncertainties in air temperature	$u_{T,i} = 0.6^\circ\text{K}$	$c_{T,i} = V_i \times V_i / (3 \times 288.15)$
Category B uncertainties air pressure	$u_{B,i} = 2.0\text{hPa}$	$c_{B,i} = c_{V,i} \times V_i / (3 \times 1013)$
Category B uncertainties relative humidity	$u_{RH,i} = 0.63\%\text{RH}$	$c_{RH,i} = c_{V,i} \times V_i \times 0.0018$

Calculation of AEP uncertainty - background

Item	Uncertainty, u	Sensitivity factor
Category B uncertainties in Electric power	$u_{P,CT,i} = \frac{0.0035}{\sqrt{3}} \cdot P_i \cdot \frac{1}{3} \cdot 3 = 0.002 \times P_i$ (class of CTs is 0.2 → the limits of the current error at 20% load is ± 0.35%, and a rectangular distribution of the error is assumed) $u_{P,VT,i} = 0 \text{ kW (no voltage transducers are used)}$ $u_{P,PT,i} = 0.005 \cdot 1285 \text{ kW} / \sqrt{3} = 3.7 \text{ kW}$ (Deiff) Class of PT is 0.5 → the limits of the power error are ± 0.5% of the range, and a rectangular distribution of the error is assumed. - Range is -220kW... 1065kW $u_{dP,i} = 0.3 \text{ kW}$ $u_{P,i} = \sqrt{((0.002 \times P_i)^2 + 3.7^2 + 0.3^2)}$	1

Calculation of AEP uncertainty - background

Category B uncertainties in wind speed	$\underline{u_{V,cal,i}} = 0.025 \text{ m/s}$ (from calibration certificate) $\underline{u_{V,class,i}} = \frac{1.3}{\sqrt{3}} \cdot (0.05 \text{ m/s} + 0.005 \cdot V_i)$ (WindSensor cup anemometer is class 1.3A, and a rectangular distribution of the error is assumed) $\underline{u_{V,mnt,i}} = 0.01 \cdot V_i$ (mounted in side boom) $\underline{u_{V,terr,i}} = 0.02 \cdot V_i$ (flat site without site calibration) $\underline{u_{dP,i}} = 0.001 \cdot V_i$ $\underline{u_{M,i}} \approx 0 \text{ m/s}$ $\underline{u_{V,i}} = \sqrt{(0.025^2 + (0.038 + 0.0038 \cdot V_i)^2 + (0.01 \cdot V_i)^2 + (0.02 \cdot V_i)^2 + (0.001 \cdot V_i)^2)}$	$\underline{c_{V,i}} = P_i - P_{i-1} / V_i - V_{i-1}$
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Calculation of AEP uncertainty - background

Category B uncertainties in air temperature	$u_{T,cal,i} = 0.1^\circ\text{K}$ $u_{T,shield,i} = 0.5^\circ\text{K}$ $u_{T,mnt,i} = 0.25^\circ\text{K}$ (estimated; sensor is at 70m) $u_{dT,i} = 0.05^\circ\text{K}$ $u_{T,i} = 0.6^\circ\text{K}$	$c_{T,i} =$ $c_{V,i} * V_i / (3 * 288.15)$
Category B uncertainties air pressure	$u_{B,cal,i} = 1.5 \text{ hPa}$ $u_{B,mnt,i} = 0.5 \text{ hPa}$ $u_{dB,i} = 1.2 \text{ hPa}$ (PT110 sensor) $u_{B,i} = 2.0 \text{ hPa}$	$c_{B,i} = c_{V,i} * V_i / (3 * 1013)$
Category B uncertainties relative humidity	$u_{B,cal,i} = 0.2 \%$ $u_{B,mnt,i} \approx 0 \%$ $u_{dB,i} = 0.6 \%$ (assumed HMP155) $u_{RH,i} = 0.63\% \text{RH}$	$c_{RH,i} = c_{V,i} * V_i * 0.0018$

Plots & tables

- Scattered plot of power P_i statistics as function of hub height wind speed V_i
- Bin-averaged power, P_i , as function of bin-averaged mean wind speed V_i including combined uncertainty as "errorbar"
- Bin-averaged C_p as function of bin-averaged mean wind speed V_i .
- Power table: bin no-i, V_i , P_i , C_p , s_i , u_i & u_{ci}
- AEP table: V_{ave} , $AEP_{measured}$, u_{AEP} (absolute), u_{AEP} (relative), $AEP_{extrapolated}$

DTU

